Rank ≥ 4 for two torsion groups

Problèmes ouverts — A. Dujella · 2026

Problem 2.4 (Dujella, 2026). Are there infinitely many elliptic curves over $\mathbb{Q}$ with the torsion group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}$ (respectively $\mathbb{Z}/8\mathbb{Z}$) and rank $\geqslant 4$?

Remark. There are infinitely many curves with rank $\geqslant 3$ and torsion groups $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}$ (Dujella & Peral (2013), Dujella, Kazalicki & Peral (2021)) and $\mathbb{Z}/8\mathbb{Z}$ (Dujella & Peral (2012), Dujella, Kazalicki & Peral (2021)).